

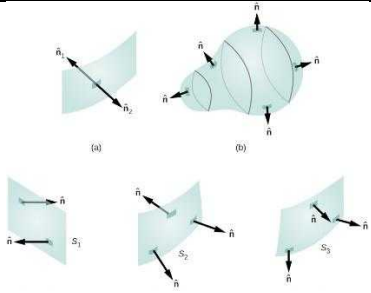
1. Dot Products (scalar product)

The dot product (also called the scalar product) is a way to multiply two vectors to produce a single scalar (number), not a vector. It's widely used in physics, machine learning, computer graphics, and linear algebra. The dot product serves a wide range of applications across various scientific and engineering disciplines, and the scalar values it produces admit diverse interpretations.

1.1 Dot Product Applications

Usually, several fields use the dot product to solve many issues during working life, such as collision detection, which can help to maintain people's lives as their assets, as shown in Table 1.

Table 1: Dot Product Application

NO	Field	Example	Shows
1	Physics & Engineering	Lifting or pushing an object along a path	
2	Electric and Magnetic Fields	Calculating the electric flux through a surface	
3	Computer Graphics	Realistic shadows, highlights, and reflections in 3D games and movies.	
4	Machine Learning	Recommender systems, search engines, and sentiment analysis	